

From d'Alembert Principle to Euler-Lagrange Equations

d'Alembert Principle $\sum_{i=1}^n \vec{F}_i^a \cdot \delta \vec{x}_i = 0$

$$\downarrow$$

$$\sum_{i=1}^n (\vec{F}_i^a - \vec{P}_i) \cdot \delta \vec{x}_i = 0$$

degree of freedom

The real d.o.f. is less than $3n$.Denote the real d.o.f. q_j , $j=1, \dots, 3n-k$, with $k = \#$ holonomic constraints.

$$\Rightarrow \delta \vec{x}_i = \sum_{j=1}^{3n-k} \frac{\partial \vec{x}_i}{\partial q_j} \delta q_j$$

$$\Rightarrow \sum_i \vec{F}_i^a \cdot \delta \vec{x}_i = \sum_i \vec{F}_i^a \cdot \frac{\partial \vec{x}_i}{\partial q_j} \delta q_j = \sum_j Q_j \delta q_j$$

where $Q_j = \sum_i \vec{F}_i^a \cdot \frac{\partial \vec{x}_i}{\partial q_j}$ is called generalized force.

Kinetic Energy $T = \sum_i \frac{1}{2} m_i \dot{\vec{x}}_i^2$,

Use two identities: $\frac{\partial \dot{\vec{x}}_i}{\partial q_j} = \frac{\partial \vec{x}_i}{\partial q_j}$, $\frac{d}{dt} \left(\frac{\partial \vec{x}_i}{\partial q_j} \right) = \frac{\partial \dot{\vec{x}}_i}{\partial q_j}$

$$\begin{aligned} \Rightarrow \sum_i \vec{P}_i \cdot \frac{\partial \vec{x}_i}{\partial q_j} &= \sum_i m_i \ddot{\vec{x}}_i \cdot \frac{\partial \vec{x}_i}{\partial q_j} \\ &= \frac{d}{dt} \left(\sum_i m_i \dot{\vec{x}}_i \cdot \frac{\partial \vec{x}_i}{\partial q_j} \right) - \sum_i m_i \dot{\vec{x}}_i \cdot \frac{d}{dt} \left(\frac{\partial \vec{x}_i}{\partial q_j} \right) \\ &= \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} \end{aligned}$$

$$\sum_i (\vec{F}_i^a - \vec{P}_i) \cdot \delta \vec{x}_i = 0$$

$$\Rightarrow \sum_{j=1}^{3n-k} \left(\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} - Q_j \right) \delta q_j = 0$$

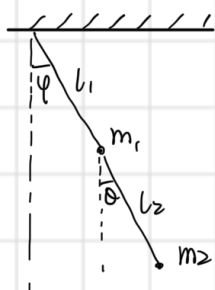
linear indep.

$$\Rightarrow \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} = Q_j, \quad j=1, \dots, 3n-k.$$

In conservative system, $Q_j = -\frac{\partial V}{\partial q_j}$ (t -independent, hence $\frac{d}{dt} = 0$).Define $L = T - V$ be the Lagrangian. $\Rightarrow$ Euler-Lagrange Equation

$$\left(\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) - \frac{\partial L}{\partial q_j} = 0 \right)$$

Ex. double pendulum

Kinetic Energy of m_1 :

$$T_1 = \frac{1}{2} m_1 l_1^2 \dot{\varphi}^2$$

$$\left. \begin{aligned} x_2 &= l_1 \sin \varphi + l_2 \sin \theta \\ y_2 &= -l_1 \cos \varphi - l_2 \cos \theta \end{aligned} \right\}$$

$$\Rightarrow v_2^2 = l_1^2 \dot{\varphi}^2 + l_2^2 \dot{\theta}^2 + 2l_1 l_2 \cos(\varphi - \theta) \dot{\varphi} \dot{\theta}$$

$$\Rightarrow T_2 = \frac{1}{2} m_2 (l_1^2 \dot{\varphi}^2 + l_2^2 \dot{\theta}^2 + 2l_1 l_2 \cos(\varphi - \theta) \dot{\varphi} \dot{\theta})$$

$$\Rightarrow T_2 = \frac{1}{2} m_2 (l_1^2 \dot{\varphi}^2 + l_2^2 \dot{\theta}^2 + 2l_1 l_2 \cos(\varphi - \theta) \dot{\varphi} \dot{\theta})$$

Gravity Potential:

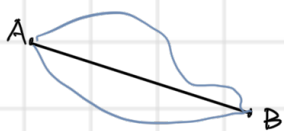
$$V_1 = -m_1 l_1 \cos \varphi$$

$$V_2 = -m_2 (l_1 \cos \varphi + l_2 \cos \theta)$$

$$\Rightarrow E = L.$$

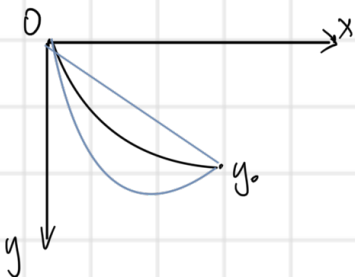
LAGRANGIAN MECHANICS FROM VARIATIONAL PRINCIPLE.

Ex1. Fermat's principle and light rays.



$$T[y] = \int \frac{ds}{v}$$

Ex2. Brachistochrone



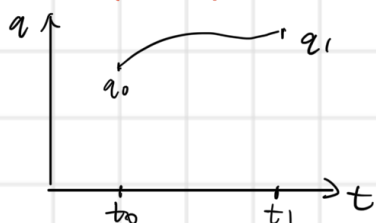
$y(x)$.

$$\frac{1}{2}mv^2 = mgy \Rightarrow v = \sqrt{2gy}.$$

$$ds = \sqrt{1+(y')^2} dx$$

$$t[y] = \int \frac{ds}{v} = \frac{1}{\sqrt{2g}} \int_0^{y_0} \sqrt{\frac{1+x^2}{y}} dx$$

Principle of least action (Hamilton's Principle)



Define an action functional

$$S[q(t)] = \int_{t_0}^{t_1} L(q(t), \dot{q}(t), t) dt.$$

$$\begin{cases} \delta S = 0 \\ q(t_0) = q_0 \\ q(t_1) = q_1. \end{cases} \xrightarrow[\text{method}]{\text{Variation}} \text{Euler-Lagrange Eqn.}$$

When there's only 1 d.o.f:

$$q_s(t) = q(t) + s\eta(t) \quad \eta(t_0) = \eta(t_1) = 0.$$

$$f(s) = S[q_s], \quad f'(0) = 0.$$

$$0 = \delta S = \int_{t_0}^{t_1} \left(\frac{\partial L}{\partial q} \eta + \frac{\partial L}{\partial \dot{q}} \dot{\eta} \right) dt$$

$$= \int_{t_0}^{t_1} \left(\frac{\partial L}{\partial q} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \right) \eta dt + \left[\frac{\partial L}{\partial \dot{q}} \eta \right]_{t_0}^{t_1} = 0$$

Since η is arbitrary,

$$= 0 \text{ everywhere} \rightarrow \text{E-L.}$$

For multi-dof, for each j we also have

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) - \frac{\partial L}{\partial q_j} = 0, \quad j=1, \dots, \# \text{dof.}$$

Elementary 1st Integrals

Define generalized momentum $P_j = \frac{\partial L}{\partial \dot{q}_j}$, if L doesn't depend on q explicitly, P_j constant, conserved quantity. first integrals.

$$\text{Energy } E = \sum_j \dot{q}_j \frac{\partial L}{\partial \dot{q}_j} - L \quad \frac{dE}{dt} = - \frac{\partial L}{\partial t}.$$

If L does not depend on t explicitly, $\frac{dE}{dt} = 0$.

One dof conservative system.

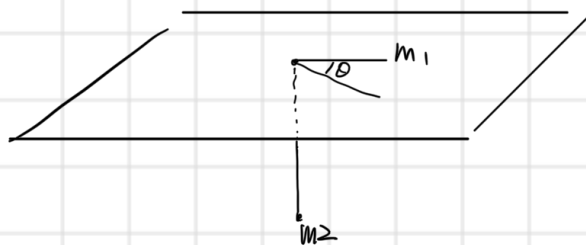
$$L = \frac{1}{2}m\dot{q}^2 - U(q)$$

$$\text{E-L equation} \Rightarrow m\ddot{q} = - \frac{dU}{dq}$$

$$E = \frac{1}{2} m \dot{q}^2 + U(q) \Rightarrow \dot{q} = \pm \sqrt{\frac{2}{m} (E - U(q))}$$

$$\Rightarrow t - t_0 = \pm \sqrt{\frac{m}{2}} \int_{q_0}^q \frac{dq}{\sqrt{E - U(q)}}$$

two d.o.f. example.



L : length of rope

$$T = \frac{1}{2} (m_1 + m_2) \dot{r}^2 + \frac{1}{2} m_1 r^2 \dot{\theta}^2$$

$$V = m_2 g (r - L)$$

$$L = T - V$$

$$\text{E-L Equation} \Rightarrow (m_1 + m_2) \ddot{r} = m_1 r \dot{\theta}^2 - m_2 g$$

$$\frac{d}{dt} (m_1 r^2 \dot{\theta}) = 0$$

define $J = m_1 r^2 \dot{\theta}$ as angular momentum which is conserved

$$\Rightarrow (m_1 + m_2) \ddot{r} = \frac{J^2}{m_1 r^3} - m_2 g. \rightarrow \text{This is an equation in an effective potential}$$

$$\downarrow \frac{dU_{\text{eff}}}{dr}$$

$$U_{\text{eff}}(r) = \frac{J^2}{2m_1 r^2} + m_2 g r$$